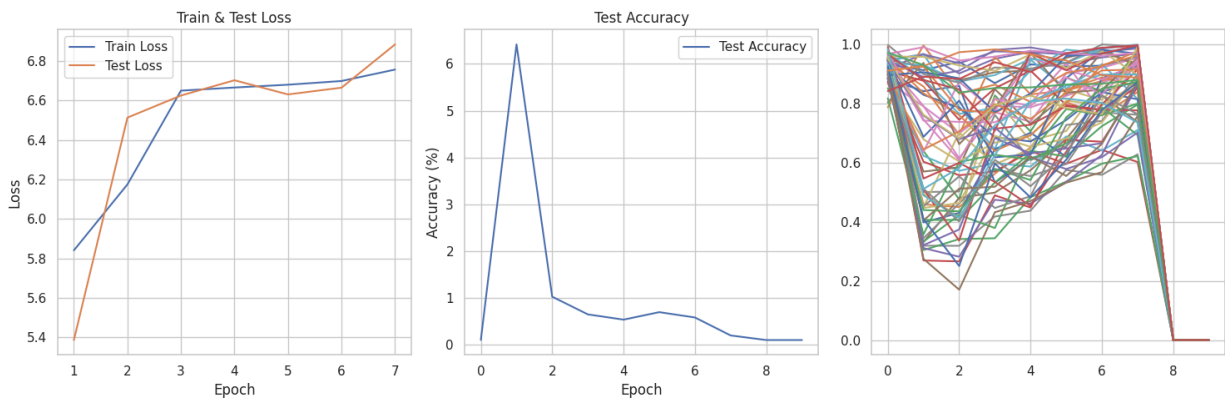


Code Overview:

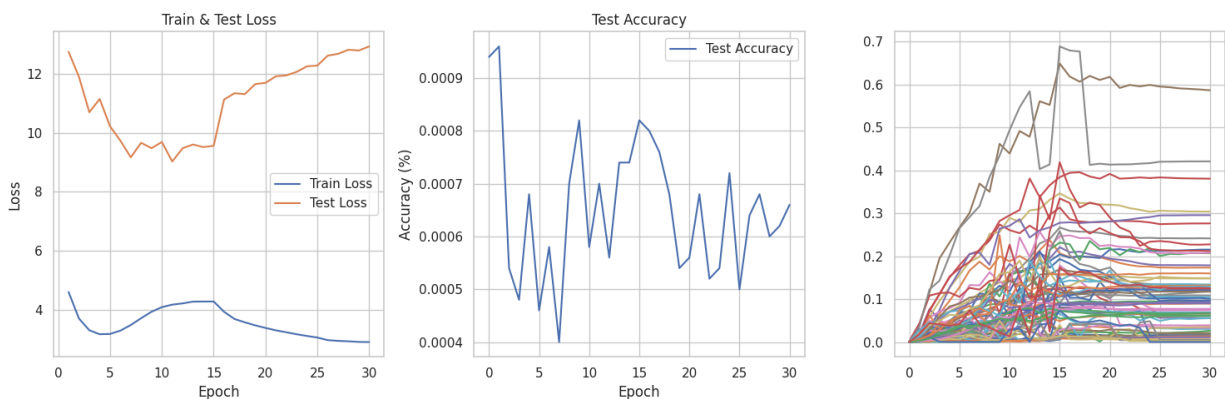
The code I implemented trains an AlexNet-style CNN on the full ImageNet dataset and examines how the filters develop in the first CNN as the training develops. Using a set of sine gratings at different degrees, we calculate the circular variance between a learned kernel and sine gratings to get a measure of directional sensitivity. Due to an error in implementation I trained the AlexNet twice. The first time I accidentally added padding = 2 to the first Conv layer, made a mistake in the accuracy calculation, and inverted the CV function. Due to limited resources, I didn't have time to fully retrain the network and had to stop at around epoch 7, well before convergence. I display and comment on both results.

Overview Plot:

Fixed (stopped at 7)



Old (finished at 30)

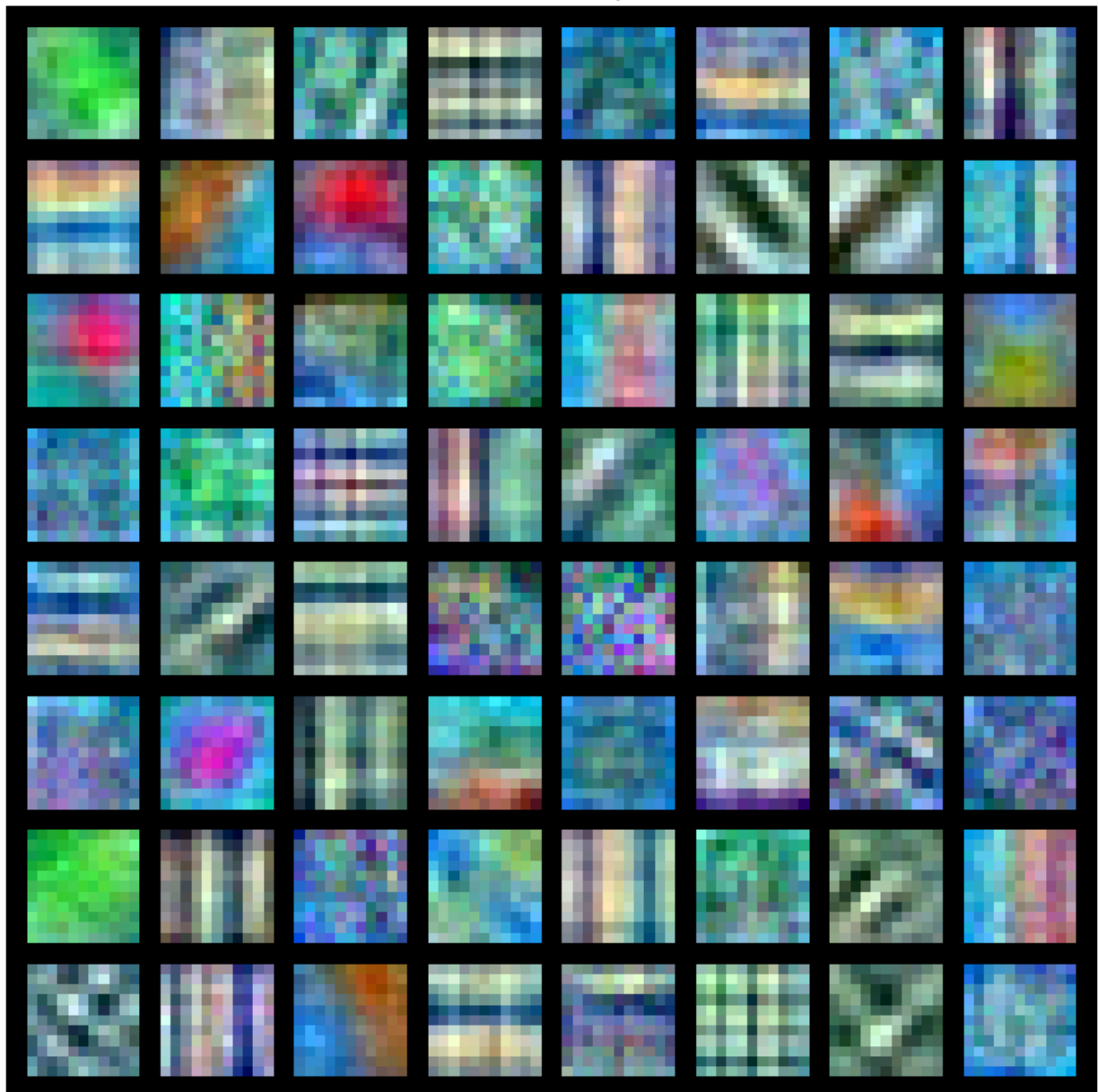


For the full model, we can see clear evidence of overfitting with a u-shaped test loss function and train loss being strictly less at all epochs. Meanwhile with the fixed model, the train less begins to diverge right at epoch 7. For CV in both plots we can see that the kernels almost immediately specialize for direction specificity. It's hard to comment on accuracy as the old plot is wrong and the new plot hasn't stabilized yet.

Kernel Viz

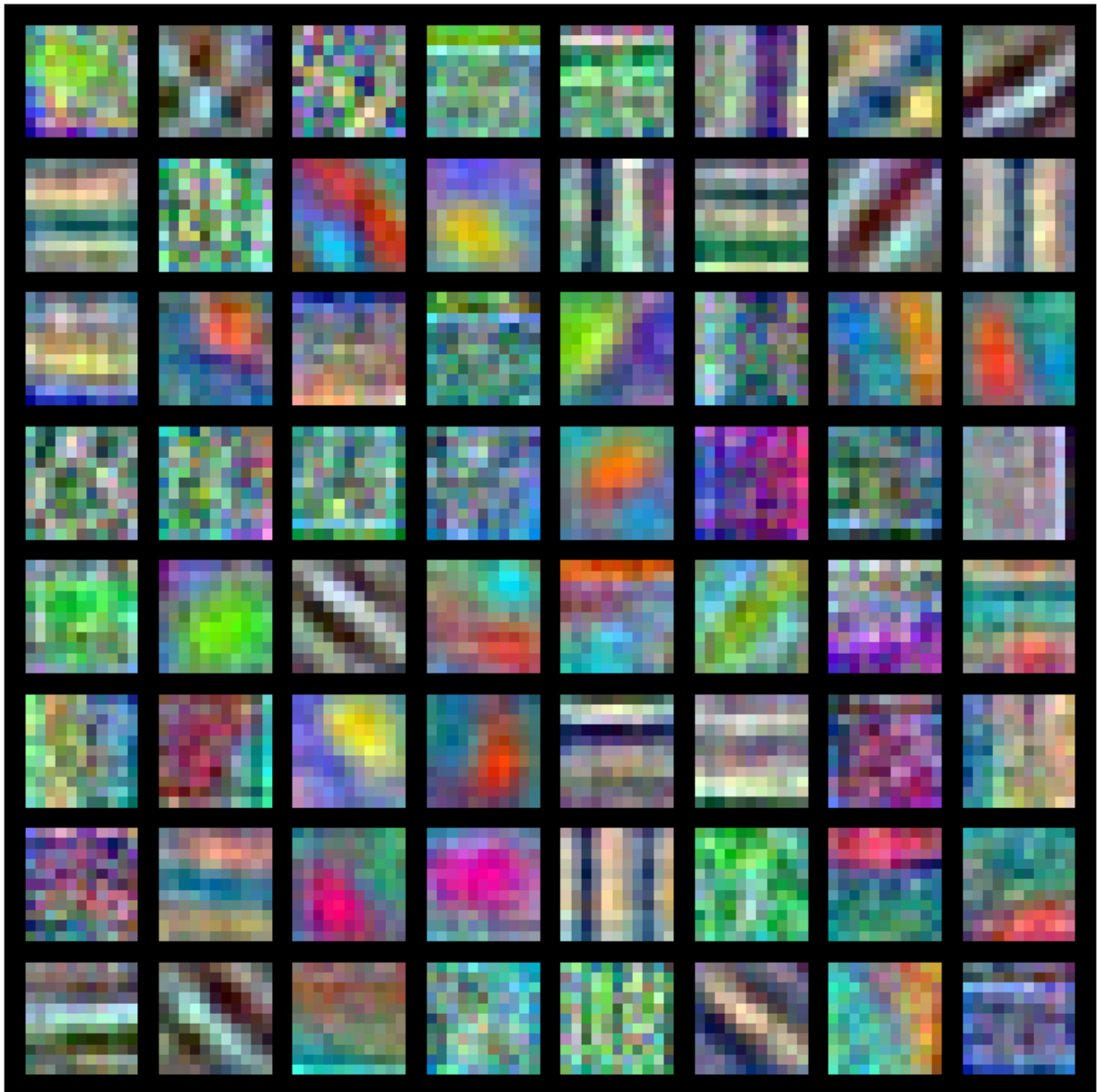
Fixed (stopped at Epoch 7)

Conv1 Kernels - Epoch 6



Old

Conv1 Kernels - Epoch 30

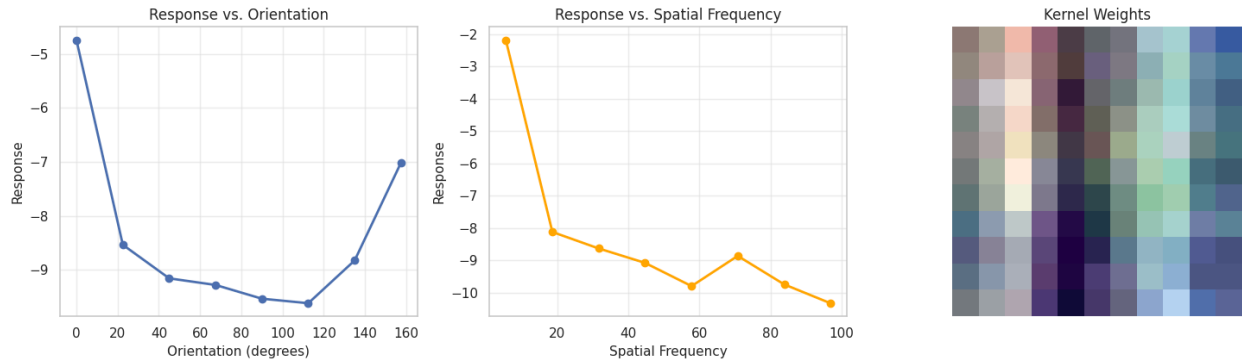


Both plots look relatively similar with some noisy kernels and some kernels that have clear directionally specificity. Looking at the fixed model we can see kernels at a variety of angles as well as checkered.

Directional Kernels

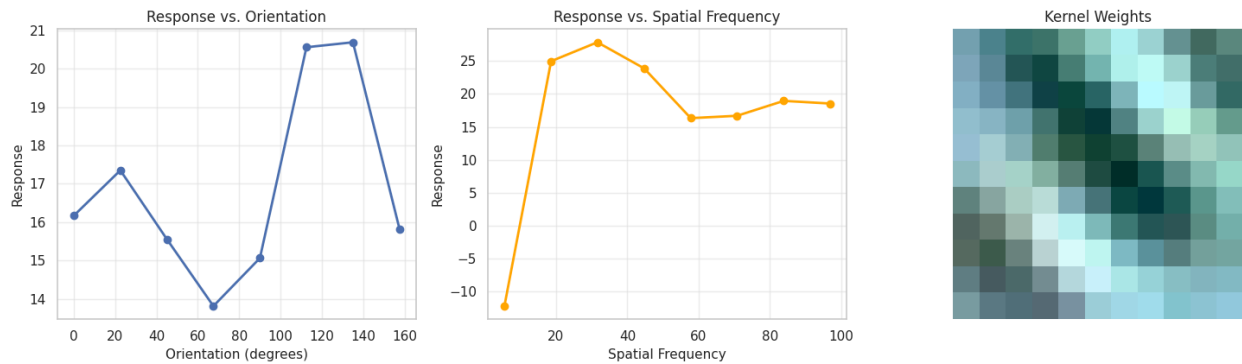
I only looked at the fixed model

Kernel 7 - Circular Variance: 0.6274



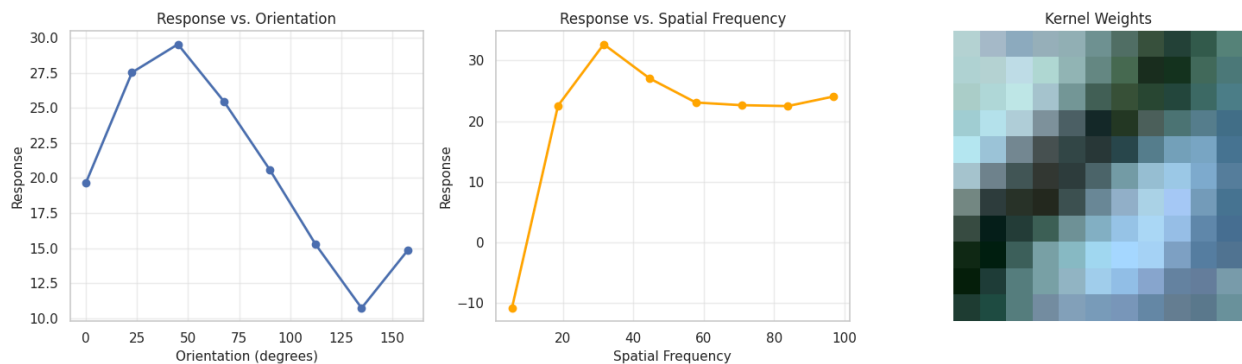
This kernel is clearly vertical with a slight slant from the top right to bottom less, this matches the response vs orientation plot. The bars are small which corresponds to a low frequency, matching the second one.

Kernel 13 - Circular Variance: 0.8550



This kernel is clearly biased in a left diagonal fashion which matches a roughly 130 degree orientation. Additionally the stripes are wide which matches a higher spatial frequency.

Kernel 14 - Circular Variance: 0.8666



Similar to the last image but rotated 90 degrees, making it sensitive to 40 degree rotations. Also responds well to higher spatial frequencies.